

GENERATIVE AI & DEEP LEARNING

Course Code: **23CS401** • Semester: **4-1** • Department of Computer Science & Engineering

Course Scope & Objectives: Deep neural network architectures, backpropagation, Convolutional Networks (CNNs), Transformers, Self-Attention, Large Language Models (LLMs), and Diffusion Models.

Unit 1: Foundations of Deep Learning & Neural Networks

Key Syllabus Topics: Perceptrons, Multi-Layer Perceptrons (MLP), Activation Functions (ReLU, Sigmoid, GeLU, Softmax), Forward Propagation, Loss Functions (Cross-Entropy, MSE), Backpropagation & Chain Rule, Optimization (SGD, Momentum, Adam, RMSprop), Regularization (Dropout, Batch Normalization, Weight Decay).

Theoretical Framework & Concepts:

Deep neural networks learn non-linear representations through hierarchical layers. The Adam optimizer dynamically computes adaptive learning rates per parameter using first and second gradient moments.

Implementation / Key Formula Reference:

```
import torch.nn as nn
model = nn.Sequential(
    nn.Linear(784, 256),
    nn.ReLU(),
    nn.Dropout(0.2),
    nn.Linear(256, 10)
)
```

Frequently Asked University Exam Questions:

• **Q1: Derive the backpropagation weight update equation using the chain rule.**

Solution: A: $dLoss/dW = (dLoss/dOutput) * (dOutput/dNetInput) * (dNetInput/dW)$, updating weights via $W = W - learning_rate * (dLoss/dW)$.

Unit 2: Convolutional Neural Networks (CNNs) & Computer Vision

Key Syllabus Topics: Convolution Operation, Strides, Padding, Pooling (Max, Average), Feature Maps, Landmark CNN Architectures (AlexNet, VGG-16, ResNet, MobileNet), Skip Connections & Residual Learning, Object Detection (YOLO, Faster R-CNN), Facial Recognition Embeddings.

Theoretical Framework & Concepts:

CNNs exploit spatial locality through weight sharing and translation invariance. ResNet skip connections resolve vanishing gradients in deep architectures, enabling models with hundreds of layers.

Implementation / Key Formula Reference:

```
conv_layer = nn.Conv2d(in_channels=3, out_channels=64, kernel_size=3, padding=1)
```

Frequently Asked University Exam Questions:

• **Q1: Why are residual skip connections critical in Deep CNNs?**

Solution: A: They allow gradients to flow unimpeded directly through the identity mapping $F(x) + x$, preventing vanishing gradients and degradation in deep networks.

Unit 3: Sequential Models, RNNs, and Attention Mechanism

Key Syllabus Topics: Recurrent Neural Networks (RNNs), Exploding/Vanishing Gradients, Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Sequence-to-Sequence Models, Bahdanau Additive Attention, Luong Multiplicative Attention.

Theoretical Framework & Concepts:

LSTMs introduce Forget, Input, and Output gates with a dedicated Cell State to maintain long-term context across lengthy sequential inputs.

Implementation / Key Formula Reference:

```
// LSTM Cell: f_t = sigma(w_f * [h_{t-1}, x_t] + b_f)
```

Frequently Asked University Exam Questions:

• **Q1: Explain the gating mechanisms inside an LSTM cell.**

Solution: A: Forget Gate decides what information to discard from cell state; Input Gate selects new candidate values to store; Output Gate produces hidden state based on filtered cell state.

Unit 4: Transformer Architecture & Self-Attention

Key Syllabus Topics: Scaled Dot-Product Attention, Multi-Head Attention, Positional Encodings, Encoder-Decoder Stacks, Feed-Forward Networks, Layer Normalization, BERT (Masked LM), GPT (Autoregressive Decoder-Only), Vision Transformers (ViT).

Theoretical Framework & Concepts:

Transformers replace recurrence with self-attention, processing all tokens concurrently. Multi-Head Attention computes $\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) * V$ across multiple subspace representations.

Implementation / Key Formula Reference:

```
Attention(Q, K, V) = softmax((Q @ K.T) / sqrt(d_k)) @ V
```

Frequently Asked University Exam Questions:

• **Q1: Explain Multi-Head Attention and why scaling by $\sqrt{d_k}$ is necessary.**

Solution: A: Scaling prevents dot products from growing excessively large for high dimensions, which would otherwise push the softmax function into regions with near-zero gradients.

Unit 5: Generative Models, LLMs, and Modern GenAI

Key Syllabus Topics: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Diffusion Models (DDPM, Stable Diffusion), Large Language Model Pretraining, Instruction Fine-Tuning (RLHF), Parameter Efficient Fine-Tuning (LoRA, QLoRA), Retrieval-Augmented Generation (RAG), Vector Databases.

Theoretical Framework & Concepts:

Generative AI creates novel synthetic text, images, and audio. RAG combines external knowledge vector retrieval with LLM generation to eliminate hallucinations and supply up-to-date domain context.

Implementation / Key Formula Reference:

```
// LoRA adaptation:  $W_{\text{new}} = W_{\text{pretrained}} + (A @ B) * (\alpha / r)$ 
```

Frequently Asked University Exam Questions:

• **Q1: Explain the architecture and workflow of Retrieval-Augmented Generation (RAG).**

Solution: A: User queries are converted to embeddings, matched against a vector database for top-k relevant document chunks, and passed as grounding context into the LLM system prompt.
